

When is one field enough?

A measurement audit of an organ-on-a-chip quality-control benchmark, a corrected evaluation protocol, and a chip-level QC tool with an explicit "inconclusive" outcome.

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Code, video, demo	https://github.com/Foreist/when-is-one-field-enough · video https://taewoong23-ooc-chip-qc-demo.static.hf.space/video.html · browser demo in §6.8

Summary

Organ-on-a-chip (OoC) cultures are inspected daily under a microscope, and the quality-control (QC) decision — keep the chip, re-image, or discard it — is still made by eye. A public benchmark exists for automating this: the **OOC Image Dataset** (3,072 brightfield fields, 59 acquisition sessions, six cell lines, labels from four expert cell biologists by majority vote). Before building on it we asked a simple question: *is this benchmark measuring what it claims to measure?*

It is not, in three specific and measurable ways.

- 1. The published split leaks sessions.** Training and test images come from the *same 57 of 59 sessions*. Holding the test images and the training-set size fixed, and changing only whether the test sessions also contribute training images, moves accuracy by **+8.2 pp (95% CI 6.0-10.4)** and **AUC by +9.6 pp (95% CI 8.4-10.8)** across eight seeds. The shipped split reports 78.9%; the same model on unseen sessions reports 56.0%.
- 2. The 3,072 labels are not 3,072 independent observations.** Labels are strongly autocorrelated along the acquisition order (session ICC 0.321, mean run length 6.08 fields versus 2.03 under independence). The design effect is 17, so the benchmark carries the information of roughly **181 independent labels** — and a single field agrees with the session's majority label only **78.8%** of the time (91.0% even with ten fields).
- 3. The failure mode is spatial, not per-image — and our own policy comparison did not survive its own criticism.** Failures occupy contiguous stretches of a chip (runs test significant in 23/45 mixed sessions, surviving a cell-type control). A policy comparison simulated on the *labels* suggests random fields are best for the chip call; **with the model in the loop that ranking disappears** — every paired difference over 13–15 chips has a bootstrap CI containing zero (§4.3). What does survive is a concrete, reproducible failure: reading the **first k** fields called a 100%-bad chip *pass* with 0.94 confidence, because the start of a session can sit inside a good region. The shipped tool samples fields spread across the chip for that reason, not on the strength of a policy ranking.
- 4. The leakage is not unique to this benchmark.** Auditing a second public organoid imaging benchmark (OCT organoid tracking, zenodo.15783866) shows the same defect:

40.0% of its test files belong to a (well, day) acquisition group that also appears in training (§4.4).

We then built the tool the corrected protocol implies, and measured what it buys: on 25 unseen chips it reaches the same chip-level accuracy as reading **every** field (0.800) while using **9.5 fields instead of 27.4** — a 2.9× reduction in microscope time — and defers 8% of chips to a human rather than guessing. Fields are sampled **spread across the chip** (reading the first k can sit inside a good region and stop early — we observed a 100%-bad chip called *pass* with 0.94 confidence), and a sequential stopping rule returns **pass / fail / inconclusive** with a posterior confidence. On 25 unseen chips: **82.6% chip-level accuracy among confident calls, 13% confident-but-wrong, 8% inconclusive, 9.5 fields on average**. A plain cap of 12 spread fields per chip does as well on field count (9.0 on average, 0.800); the saving comes from not reading every field, and what the sequential rule adds is a posterior confidence and the option to defer.

We also report two things we could not do, because they define the honest boundary of this work: the **re-image vs discard** distinction is *not* supported by the data (a constructed recovery target is not predictable; CV AUC 0.613 against a permutation null of 0.564), and **no reliable out-of-distribution detector** was found (image statistics and feature-space distance both missed the worst failure). Finally, simulating a stopping rule on ground-truth labels **overstates** it: on the same 25 chips, without a minimum-field guard, labels give 0.875 and the model 0.708.

Everything is reproducible from one public dataset with the scripts in this repository.

1. Introduction

1.1 The problem

Microphysiological systems — organ-on-a-chip, organoids in microfluidic devices — are promoted as human-relevant alternatives to animal testing. Their practical bottleneck is not the biology but the *bookkeeping*: a chip is imaged repeatedly over days, and at each session someone must decide whether the field in front of them is usable. Technical artifacts (air bubbles, defocus, deformed channel walls) and biological problems (cells not attaching, density far from the expected range) both appear as "bad", and both are judged by eye, field by field.

Automating this judgement is attractive, and a public benchmark for it exists. But an automated QC system inherits every property of its benchmark: if the benchmark's split leaks, reported accuracy is inflated; if its labels are clustered, its effective sample size is far smaller than it looks; if its labels are field-dependent, then "accuracy per image" is not the quantity a lab actually needs. None of these properties are usually checked.

Who this is for. (i) *Chip and organoid labs* that image cultures daily and decide keep / re-image / discard by eye; (ii) *imaging and data teams* building reusable, trustworthy culture datasets; (iii) *assays downstream of QC* — dose-response, toxicity, drug evaluation — which inherit the quality of the fields that were kept.

1.2 What this report does

We take one public OoC QC benchmark — the OOC Image Dataset [1,2] — and treat the benchmark itself as the object of study. We (i) audit its split and label structure, (ii) derive and validate a corrected evaluation protocol, (iii) measure how sampling and aggregation policies behave on the measured structure, and (iv) build and honestly evaluate a chip-level QC tool.

1.3 What we claim — and what we do not

We claim **measurement results** about a specific benchmark and a **protocol/tool** that follows from them. We do *not* claim a new model architecture, biological or clinical validity, or generalisation beyond this dataset. Leakage and label clustering are known phenomena in machine learning [7,8,9,10,11]; our contribution is not their discovery but their **measured magnitude in this benchmark, the corrected protocol, and the field-sampling policy that the measured structure implies** — together with two negative results that bound the claims.

2. Data

Source. OOC Image Dataset, Zenodo record 10.5281/zenodo.10203721 [3]; data descriptor in *Data* 2024 [1] and a companion conference paper [2]. The Zenodo record states **CC-BY-4.0**; the MDPI descriptor lists **CC-BY-SA**. We resolve the ambiguity conservatively: we redistribute no images beyond the 44 attributed demo example fields, license our code MIT and our derived artefacts CC-BY-SA.

Content. 3,072 brightfield images ($2,056 \times 1,542$ px) from an automated microscope on an OoC setup, covering six cell lines (A549, Caco-2, HPMEC, HUVEC, NHBE, HSAEC). Filenames are `YYMMDD_N.png`: `YYMMDD` identifies an **acquisition session** (59 in total) and `N` the field number within it. Each image carries a binary label, `good` or `bad`, assigned by **four expert cell biologists**, with disagreement resolved by majority vote or an additional rater [1]. Metadata (cell type, seeding density, flow rate, day after seeding) is available for part of the data. Figure 1 summarises the corpus: sessions range from 3 to 229 fields (median 24), and the share of fields labelled `bad` ranges from 0% to 100% per session.

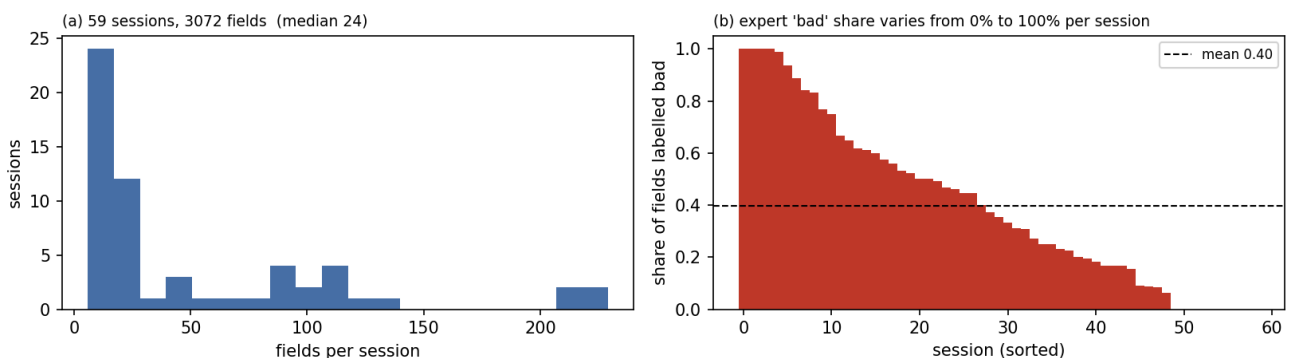


Figure 1. The corpus. (a) fields per session; (b) per-session share of fields labelled `bad`.

The published split. The zip ships `train/val/test` folders (2,130 / 286 / 656 images). This is the split used by the reference baseline in [1].

Privacy, ethics and compliance. The dataset contains brightfield images of immortalised cell lines grown in microfluidic devices. There are no human subjects, no patient-derived material, no identifiable personal data and no clinical records, so no consent or IRB review applies; the data are distributed publicly under the licences above. The tool automates a quality judgement about an image — it makes no biological, diagnostic or clinical claim, and we do not use it to make decisions about patient material. The second benchmark audited in §4.4 is likewise public (CC-BY-4.0) and contains only instrument images.

3. Methods

All quantities below are defined once here and used unchanged throughout; the scripts that compute them are named in §7.

Sessions and fields. A *session* is one acquisition date (YYMMDD in the filename) — one microscope run over one or more chips. A *field* is one image. We treat the field index N as the acquisition order; the proxy is supported by the high correlation between consecutive fields (§4.2). A *run* is a maximal contiguous block of equal labels in field order.

Runs test (clustering). For a binary sequence with n_1 bad and n_0 good fields, the number of runs R has, under independence, mean $\mu = 2n_1n_0/n + 1$ and variance $\sigma^2 = 2n_1n_0(2n_1n_0 - n) / (n^2(n - 1))$; we report $z = (R - \mu)/\sigma$ and the two-sided normal p-value. Negative z means fewer runs than chance, i.e. clustering. Sessions without both labels are skipped. The i.i.d. expected run length is $1 / (2p(1-p))$ with p the observed good fraction.

Intraclass correlation and design effect. For clusters (sessions) with sizes n_i , ICC(1) is estimated by one-way ANOVA, $ICC = (MSB - MSW) / (MSB + (m_0 - 1) \cdot MSW)$ with $m_0 = (N - \sum n_i^2/N)/(k - 1)$; the design effect is $1 + (m_0 - 1) \cdot ICC$ and the effective sample size is $N / \text{design effect}$. This is the standard cluster-sampling treatment of "images within a chip".

Controlled leakage A/B. Test images are fixed by choosing 20 sessions and withholding half of each session's fields as the test set. The other half of those sessions' fields are *available* but used only in the leaky arm. Both arms receive exactly N training fields: the disjoint arm draws all N from non-test sessions; the leaky arm draws $N - k$ from non-test sessions and k from the available test-session fields. Model, budget, augmentation and seeds are identical.

Chip-level decision. For a set of fields with per-field bad probabilities p_j , the sequential rule maintains a Beta(1,1) posterior on the bad fraction f — $f \mid \text{data} \sim \text{Beta}(1 + b, 1 + g)$ with b, g the counts of fields with $p_j > 0.5$ and ≤ 0.5 — and stops as soon as $P(f > 0.5) > 0.9$ or $P(f < 0.5) > 0.9$, at most after a budget of 20 fields and never before `min_fields`. If the budget is exhausted with $0.35 < P(f > 0.5) < 0.65$, the call is **inconclusive** rather than forced.

Spread-field order. With n available fields and a budget of k , the rule inspects indices `round(i · (n-1)/(k-1))` for $i = 0 \dots k-1$, i.e. fields spaced evenly across the acquisition order. This replaces "the first k fields" after the failure described in §6.2.

Metrics. *Field accuracy* is the fraction of fields whose predicted side of 0.5 matches the expert label. *Chip reference* is the session's majority label. *Chip accuracy among confident calls* counts only calls the rule made with posterior confidence ≥ 0.9 ; *false-confident rate* is the fraction of all chips that received a confident but wrong call; *inconclusive rate* is the fraction that exhausted the

budget near 0.5. *Bad-region recall* (sampling simulations) is the fraction of the session's bad fields that the sampled set contains.

Evaluation discipline. Every model number in this report comes from a **session-disjoint** test set of 25 sessions (684 fields). Where we also report the shipped split's number, it is labelled as such. Hyperparameters were fixed before the final evaluation; no test-set tuning was performed.

4. Audit

4.1 The published split leaks sessions

Observation. Parsing the archive's central directory shows the split is image-level: sessions are not held out. **Train contains 59 sessions, validation 51, test 57, and train $\cap$ test = 57 of 59 sessions** — images of the same chip, acquired in the same session, appear on both sides.

Controlled measurement. A plain comparison of splits confounds leakage with differences in test composition, so we hold everything else fixed:

- the **test images** are identical in both arms (20 sessions, 521–712 fields);
- the **training-set size is identical** in the two arms (1,440–1,823 fields depending on the seed);
- the **model and budget are identical** (MobileNetV3-small [4], ImageNet-initialised, 6 epochs, 224 px);
- the only difference is whether the training set may use *the other images of the test sessions* (leaky) or must come from disjoint sessions (disjoint).

arm	accuracy	AUC
session-disjoint	67.2% (± 6.2)	0.750
same sessions in training (leaky)	75.4% (± 6.8)	0.846
inflation (paired, 8 seeds)	+8.2 pp [6.0, 10.4]	+9.6 pp [8.4, 10.8]

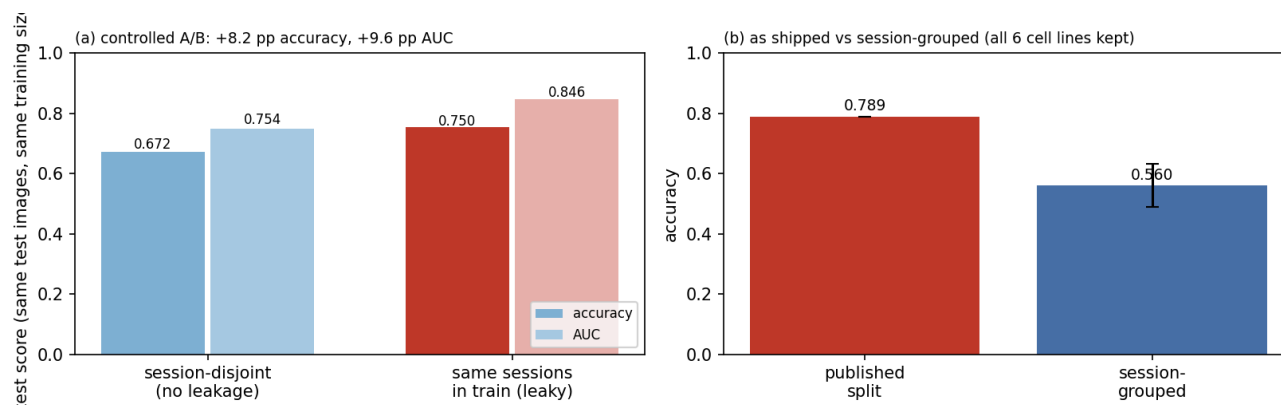


Figure 2. (a) controlled A/B (8 seeds; paired inflation +8.2 pp accuracy, +9.6 pp AUC); (b) the shipped split versus a session-grouped split.

For reference, the shipped split yields **78.9% $\pm$ 0.07 / AUC 0.873** (seed-to-seed variation is tiny because all test sessions are seen in training), while a session-grouped split that keeps all six cell lines yields **56.0% $\pm$ 7.2 / AUC 0.684** — a 22.9 pp gap that mixes leakage with the small

number of held-out sessions (six). We therefore treat the **controlled A/B as the headline number** and report the shipped/grouped pair only as context.

Why this matters for the benchmark. Any model compared on the shipped split is compared on a test set whose sessions it has already seen. Model *rankings* may survive (the leak affects all models similarly), but the *absolute* numbers, and any claim of "generalisation to new chips", do not.

4.2 Labels are clustered along the acquisition order

Runs test. For each session we take the label sequence in field order and count runs (maximal contiguous blocks of equal label). Under independence the expected run length is $1 / (2 p (1-p))$; with $p \approx 0.56$ that is **2.03** fields. Observed: **6.08**. A Wald-Wolfowitz test is significant ($p < 0.05$) in **23 of the 45 mixed sessions**, and 36/45 have negative z (clustered) — against 22.5 expected by chance. The most extreme session has $z = -10.3$ (27 runs where 82.5 were expected, 229 fields).

session	fields	good share	runs	expected runs	z	p
221010	99	0.82	13	30.5	-5.97	<1e-6
230109	107	0.80	14	34.8	-6.43	<1e-6
230314	50	0.44	3	25.6	-6.57	<1e-6
230315	94	0.55	13	47.5	-7.23	<1e-6
230317	100	0.73	10	40.4	-7.78	<1e-6
230320	229	0.77	27	82.5	-10.34	<1e-6
230419	124	0.07	2	16.0	-10.71	<1e-6
230424	108	0.51	21	55.0	-6.57	<1e-6

Table 1. The eight most clustered sessions (all $p < 1e-6$). Expected runs are what independence would give for the same good/bad counts.

This is not a cell-type artifact. A session can contain several cell types imaged in contiguous blocks, which would create clustering for trivial reasons. Restricting the test to *maximal contiguous same-cell-type stretches*, **38 of 72 stretches remain significant** (57/72 clustered).

It is not duplicate frames either. Downscaled correlation between consecutive fields is 0.63 on average (random pairs: 0.09–0.38), i.e. consecutive fields are spatially adjacent — but a greedy view-clustering at $r > 0.95$ finds **1,161 distinct views among 1,183 images (98%)**. The clustering is therefore not redundancy; **failures occupy contiguous regions of the chip**.

Effective sample size. With a session-level intraclass correlation of **ICC = 0.321** and a mean cluster size of ~ 51 fields, the design effect is $1 + (m-1) \cdot \text{ICC} = 17.0$, so for estimating a metric the benchmark carries the information of roughly **3,072 / 17 $\approx$ 181 independent labels** (a model may still exploit correlated fields in ways an estimator cannot, so this is a bound on *estimation*, not on learning). The mean lag-1 autocorrelation is 0.32 (53% of sessions above 0.3).



Figure 3. (a) runs-test z per session; (b) observed versus i.i.d. run length; (c) ICC, design effect and effective sample size.

Agreement degrades with culture age and is cell-line dependent:

culture age	fields	single-field agreement
0-1_days	852	0.805
2-3_days	798	0.779
4+_days	1199	0.676
4_days	223	0.704

cell line	fields	single-field agreement
A549	775	0.710
CACO	346	0.792
HPMEC	1462	0.702
HSAEC	244	0.840
HUVEC	107	0.991
NHBE	138	0.826

Table 2. Single-field agreement with the session's majority label, by culture age and cell line. Late cultures are the hardest, and no cell line is uniformly easy.

Consequence for labels, not just for statistics. If the label were a property of the chip, a single field would predict the session's majority label almost perfectly. It does not: a single field agrees with the session majority **78.8%** of the time, three fields 84.1%, five 87.4%, and even ten fields only **91.0%**. Agreement is worse for late cultures (4+ days: 67.6% for a single field) than for day 0–1 (80.5%). The "quality" of a chip is therefore **a property of a region and of the sampling**, not of an image.

4.3 Sampling policy: a comparison that does not survive the model in the loop

If failures are contiguous, how should fields be chosen? We first simulate three policies on the measured **label sequences** at a fixed budget k (300 trials per session):

- **random** — k distinct uniform fields;
- **window** — one contiguous run of k fields (a scan);

- **adaptive** — start at a random field; if it was bad, expand to an unsampled neighbour; if good, jump elsewhere.

budget k	random: acc / recall / err	window: acc / recall / err	adaptive: acc / recall / err
2	0.822 / 0.095 / 0.182	0.812 / 0.096 / 0.222	0.808 / 0.105 / 0.206
4	0.860 / 0.196 / 0.119	0.833 / 0.188 / 0.174	0.838 / 0.214 / 0.153
8	0.907 / 0.351 / 0.074	0.886 / 0.328 / 0.129	0.851 / 0.391 / 0.120
12	0.920 / 0.429 / 0.055	0.891 / 0.413 / 0.106	0.849 / 0.474 / 0.110
20	0.922 / 0.402 / 0.046	0.876 / 0.380 / 0.107	0.789 / 0.482 / 0.137

Table 3. Label-based policy simulation. *err* = |estimated – true bad fraction|. Random looks best for the call and adaptive best for localisation — and we do not believe this table.

This is exactly the practice criticised in §6.5, so we repeat the comparison with the deployed model's per-field probabilities on the 25 session-disjoint test chips (`audit/03c_policy_model_in_loop.py`):

budget k	random: acc / recall	window: acc / recall	adaptive: acc / recall
4	0.791 / 0.335	0.785 / 0.321	0.772 / 0.323
8	0.789 / 0.300	0.818 / 0.311	0.776 / 0.296
12	0.800 / 0.393	0.818 / 0.389	0.840 / 0.397

Table 4. The same policies with the model in the loop. The ordering changes, and no difference is significant: a paired bootstrap over chips (2,000 resamples) gives 95% intervals of $[-0.08, +0.00]$ for random – adaptive at $k=12$ (13 chips), $[-0.12, +0.07]$ for window – adaptive and $[-0.11, +0.07]$ for random – window — all contain zero (Figure 4c).

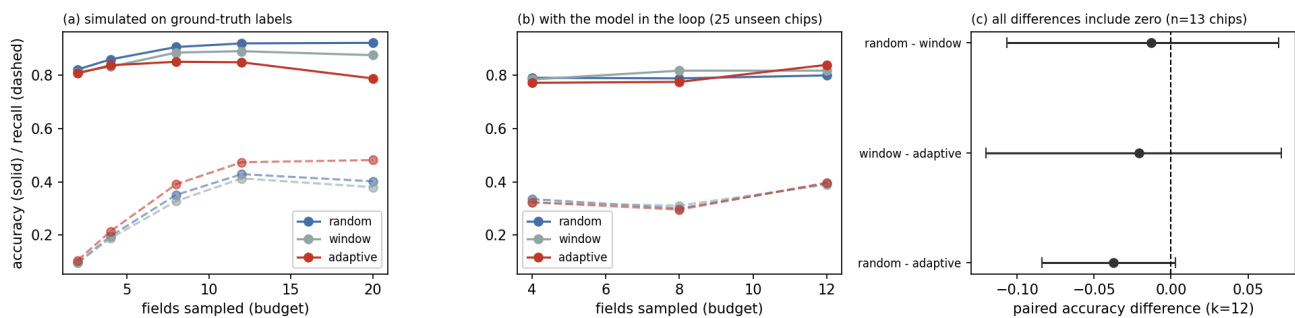


Figure 4. Policy comparison. (a) simulated on ground-truth labels; (b) with the model in the loop; (c) paired bootstrap differences at $k=12$ — all intervals include zero, so no ranking is claimed.

We therefore report no policy ranking. With only 13–15 chips large enough for these budgets the comparison is underpowered, and the label-based ranking is an artefact of scoring policies against the very labels they were designed to sample.

What *is* established, and what the tool uses, is a single reproducible failure of the obvious default. Reading the **first** k fields in acquisition order with the sequential rule of §6.1 called a 100%-bad chip (session 230405) *pass* with 0.94 confidence: the per-field probabilities in the first third of that session are low (mean 0.255 over the first five fields, versus 0.935 over the last twenty). Sampling fields spread across the chip fixes that case (the same chip returns *fail*, $P = 0.91$); on

the 11 chips with at least 20 fields the accuracy difference (0.727 vs 0.636) is ± 1 chip and is not significant either. The justification for spread sampling is the observed failure, not a measured ranking.

4.4 The same defect in a second organoid benchmark

To test whether the leakage we measure is a property of one dataset or of the field, we audited a second public organoid imaging benchmark: the **OCT organoid segmentation-and-tracking dataset** (zenodo.15783866, CC-BY-4.0; Branciforti et al., *Diagnostics* 2024 [5,6]). Its file names encode the acquisition group — `w<well>_d<day>_<slice>.png` in train/val and `d<day>_p<plate>_w<well>_<slice>.png` in test — and the same (well, day) means the same organoids imaged in the same session. Reading the archive's central directory over HTTP range requests (no 4.9 GB download; `audit/06_oct_leakage.py`):

- train: 16,752 files in **8** (well, day) groups; val: 4,188 files in 2 groups; test: 20,940 files in 10 groups
- **train n test = 4 groups** — well 2 at days 5, 7, 11 and 13; val n test = 1 group
- **40.0% of test files (8,376 of 20,940) belong to a (well, day) group that also appears in training**

For a *tracking* benchmark this is more severe than for classification: the same organoid instances, imaged in the same session at the same timepoint, appear on both sides of the split. We report the structural overlap rather than a re-trained inflation number, because re-training their segmentation/tracking pipeline is out of scope here.

Two datasets, two independent instances of the same defect — group-level splits that leak — which is the empirical basis for the first rule of the corrected protocol (§5).

4.5 A hypothesis we tested and rejected: re-image vs discard

The dataset's own definition of bad mixes **technical artifacts** (bubbles, defocus, deformed walls) with **biological problems** (morphology/density off expectation) [1]. If these could be separated, the actionable output would be "re-image" (cheap, the chip survives) versus "discard" (expensive, re-imaging is useless). Two observations encouraged the attempt: isolated bad runs are blurrier and darker than good fields (Laplacian 0.089 vs 0.124; dark fraction 0.137 vs 0.091 — bubbles and defocus), whereas long bad runs have normal focus but higher contrast; and visually, 4/4 sampled isolated-bad fields showed bubbles or defocus.

We constructed a target — a run "recovers" if the next five fields are majority good — and tried to predict it from features available at decision time (run length, whether the run reaches the end of the session, focus, darkness, artifact scores), with **session-grouped cross-validation**.

feature set	CV AUC
run length only	0.476
artifact signals only	0.381
run length + artifacts	0.539
all features	0.613
permutation null	mean 0.479, p95 0.564
simple rule ($\text{length} \leq 2 \Rightarrow \text{recover}$)	accuracy 0.560 vs base rate 0.624

The decision rule is not supported. The target is confounded by position (because failures are contiguous, the fields after a run are usually good, so "recovery" largely means "the run did not reach the end"), and — more fundamentally — **no re-imaging was ever performed in this dataset, so the causal question cannot be observed.** We therefore ship `pass/fail/inconclusive` and document the two-mode observation as an *open problem*, not a feature.

4.6 Aggregation rules are a sensitivity/specificity dial

If several fields are read, their scores must be combined. We compare three rules on per-field probabilities from a session-disjoint model, at a fixed field threshold chosen on validation, with the chip reference defined as before:

budget k	mean: acc / sens / spec	max: acc / sens / spec	top-2: acc / sens / spec
1	0.704 / 0.456 / 0.832	0.704 / 0.456 / 0.832	0.704 / 0.456 / 0.832
3	0.691 / 0.475 / 0.803	0.699 / 0.691 / 0.704	0.709 / 0.610 / 0.760
5	0.678 / 0.493 / 0.767	0.626 / 0.686 / 0.598	0.627 / 0.574 / 0.653
8	0.742 / 0.645 / 0.787	0.627 / 0.878 / 0.507	0.660 / 0.787 / 0.598
12	0.742 / 0.851 / 0.710	0.570 / 1.000 / 0.354	0.631 / 0.997 / 0.452

Table 5. Aggregation rules (224 px model, 2 seeds × 25 held-out sessions; 60 random k-field draws per chip, shared by all rules; chips with < k fields excluded).

`mean` maximises chip accuracy and keeps specificity high; `max` maximises sensitivity (it calls almost everything bad: specificity collapses to 0.35 at k=12) and `top-2` sits between. **The aggregation rule moves the operating point more than the per-field model does** — at k=12, accuracy ranges from 0.570 to 0.742 and sensitivity from 0.851 to 1.000 depending only on the rule. A QC deployment must therefore state its aggregation rule; "accuracy" without it is not a specification.

We also tested a rule that exploits the measured autocorrelation — smoothing the per-field scores along the acquisition order before aggregating. It did **not** help: on the shipped predictions a 3- or 5-field moving average changes chip accuracy by at most 0.007 at any budget from 4 to 12 fields (k=6: 0.792 → 0.793; `audit/smoothing_test.py`). Reported as a negative result.

4.7 Metadata does not explain the labels

A logistic model on day, log seeding density, flow rate and cell line reaches only 0.562 in-sample accuracy, and apparent cell-line differences (Caco-2 68.5% bad vs NHBE 23.9%) dissolve on inspection: every cell line has sessions with a 0% bad rate and sessions with 100%. The label is dominated by **session-level** variation, consistent with §4.2.

5. Corrected evaluation protocol

From the audit, three rules follow for anyone training or benchmarking on this dataset:

1. **Split by session, not by image.** No session may contribute to two splits. Report the number of held-out sessions alongside the metric — with 59 sessions, a 20% held-out split is ~12 sessions and the confidence interval on chip-level accuracy is correspondingly wide.
2. **Evaluate at the chip level.** Report (i) per-field accuracy/AUC *and* (ii) chip-level accuracy with the number of fields used, because per-field numbers do not answer the lab's question and are inflated by within-session correlation (effective $N \approx 181$, not 3,072).
3. **State the sampling policy.** Which fields were used, and how many. Policies differ in principle (§4.3) and we could not rank them with this benchmark's 59 sessions; a number is only interpretable together with the policy that produced it.

As a worked example, `audit/01_split_audit.py` and `evaluate.py` reproduce every number in this report from the raw dataset.

6. The tool

6.1 Design

`inference.py` takes the fields of one chip (filenames in acquisition order) and returns:

- **per-field P(bad)** from a MobileNetV3-small (ImageNet-initialised, 384 px, 25 epochs, cosine LR, label smoothing, random-resized-crop/flip augmentation), trained on a **session-disjoint** split (1,495 train / 200 val / 684 test fields from 25 unseen sessions);
- a **chip call** — `pass`, `fail`, or `inconclusive` — from a Beta(1,1) posterior on the bad fraction, stopping when $P(\text{chip bad}) > 0.9$ or < 0.1 ;
- **how many fields were needed**, with fields chosen **spread across the chip** rather than the first k ;
- a **QC map** (field index vs $P(\text{bad})$) and a `chip_report.json`.

Model and training. MobileNetV3-small [4], ImageNet-initialised, classifier head replaced with a 2-way linear layer. Input 384×384 (the source images are $2,056 \times 1,542$). AdamW (lr $3e-4$, weight decay 0.02), cosine schedule to lr/30, label smoothing 0.05, batch 32, 25 epochs (≈ 14 s/epoch on one RTX 3090, ≈ 6 min total), RandomResizedCrop(0.7–1.0) + horizontal/vertical flips. Label smoothing and the modest capacity are deliberate: the labels are four-rater majorities, and 3,072 fields with an effective N of ~ 181 do not support a large model. The checkpoint is shipped in `model/`.

6.2 Two bugs found during deployment (and why they matter)

Deploying the tool on real chips exposed two failure modes that a label-only simulation had hidden.

(a) Reading the first k fields is wrong. On a 100%-bad chip, walking the first fields in acquisition order stopped after six fields (five good predictions, one bad) and returned *pass* with 0.94 confidence. Because failures are contiguous (§4.2), the beginning of a session can lie inside a good region. Sampling fields spread across the chip fixed this specific case (the same chip now returns *fail* at 0.91).

(b) The Beta posterior is overconfident on few fields. With three good fields and no bad one, $P(\text{chip bad}) = 0.0625$ — "94% confident" — which is indefensible when the per-field model

is imperfect. Requiring a minimum of eight fields before a stop is allowed cut the false-confident rate from 25% to 13%.

6.3 Measured performance (25 unseen chips, 684 fields)

metric	value
per-field accuracy / AUC	0.734 / 0.791
chip accuracy, all fields, mean	0.80
chip accuracy among confident calls (spread fields, min 8, conf 0.9)	0.826 (95% Wilson CI 0.63–0.93, n=23)
false-confident calls (confident and wrong)	13%
inconclusive (budget exhausted near $P = 0.5$)	8%
mean fields used	9.5 (a plain cap of 12 spread fields: 9.0 fields, 0.80)

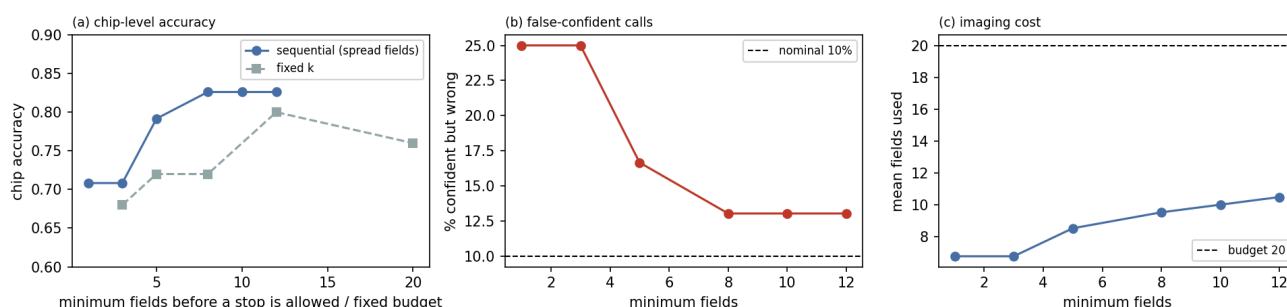


Figure 5. Tool behaviour versus the minimum-fields guard.

Capacity is not the bottleneck. We trained a backbone with 2.8× the parameters (MobileNetV3-large, same 384 px, same protocol). Test field accuracy moved from 0.734 to 0.746 and balanced accuracy from 0.733 to 0.747, while AUC *fell* from 0.791 to 0.788 — i.e. a larger model buys nothing here (audit/perfield_model.py --arch large). Together with the session-level domain shift in §6.4 this suggests the ceiling is set by the labels and by chip-to-chip appearance, not by model capacity. We therefore ship the smaller model, which is cheaper to run; with the same stopping rule the larger model uses the same 9.5 fields and is not more accurate on the chips it calls (0.792 vs 0.826).

6.4 Field efficiency: the same accuracy with 2.9× fewer fields

The naive way to use a per-field model is to read every field of the chip and average. On the 25 test chips that costs **27.4 fields per chip for 0.800 accuracy**. The shipped rule — fields spread across the chip, Beta posterior, stop at confidence 0.9, minimum 8 fields — reaches **0.800 on all 25 chips using 9.5 fields (2.9× fewer)**, and when it is allowed to defer the chips it cannot resolve it reaches **0.826 on the 23 it calls, deferring 8%** (audit/07_efficiency.py):

policy	fields per chip	chips called	accuracy
all fields (mean of every field)	27.4	25/25	0.800
fixed k = 12, spread (chips with ≥ 12 fields only)	12.0	13/25	0.846
cap of 12 spread fields (all fields if fewer)	9.0	25/25	0.800
sequential, force (min 8)	9.5	25/25	0.800
sequential, abstain (min 8)	9.5	23/25	0.826

Table 6. Field efficiency. The fixed-k row is evaluated only on the 13 chips that have at least 12 fields, so it is not comparable with the others; the cap and sequential rows use every chip. A plain cap of 12 fields is as frugal as the sequential rule, but the right cap is only known after the fact (a cap of 8 gives 0.720, a cap of 20 gives 0.760); the rule's contribution is the calibrated pass / fail / inconclusive call and the option to defer, not a further cut in fields.

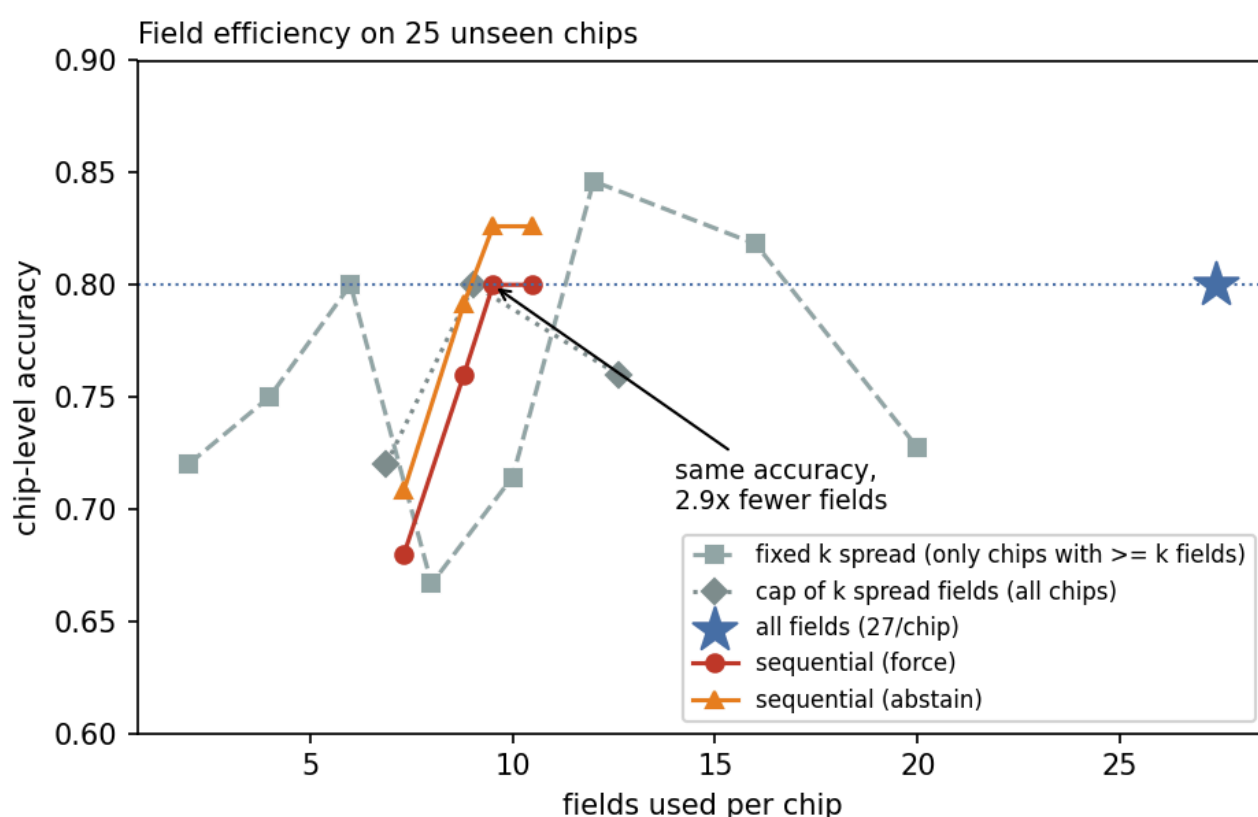


Figure 6. Accuracy against the number of fields used per chip.

For a lab imaging plates of chips this is the practical number: the decision costs roughly a third of the microscope time of the read-everything workflow, and a further 8% of chips are flagged "needs a human" instead of being guessed.

The tool ships this as a **plate mode** (`inference.py --plate <folder>`; one subfolder per chip), which returns a ranked triage table. On the 25 test chips it spends **238 of 684 fields (65% saved)** and returns 6 fail, 2 inconclusive and 17 pass, worst first.

6.5 Where the tool fails

Chip-level failures are not spread evenly; they cluster in a few sessions, and they are not flagged by the model's own confidence. The clearest case is session 230405 (69 fields, **100%** labelled

bad): the model is right on 82.6% of its fields individually, yet the per-field probabilities in the first third of the session are low (mean 0.255 over the first five fields versus 0.935 over the last twenty), so an unguarded rule stopped early with a confident *pass*.

chip (session)	fields	true bad share	per-field accuracy
220706	5	100%	0.200
220718	4	100%	0.750
220721	3	100%	1.000
230405	69	100%	0.826
230403	6	83%	0.167
230314	25	64%	0.720
230512	44	61%	0.795
230529	108	59%	0.769
230523	55	55%	0.582
220606	10	50%	0.200
230425	107	49%	0.673
230315	47	47%	0.638
230215	12	42%	0.583
220506	7	29%	0.857
220715	7	29%	0.429
230123	13	23%	0.692
220515	6	17%	0.833
230109	53	13%	0.792
230214	24	12%	0.667
220502	4	0%	1.000
220620	6	0%	1.000
220716	4	0%	0.750
230119	35	0%	0.971
230316	22	0%	1.000
230321	8	0%	1.000

Table 7. The 25 unseen test chips, sorted by the true share of bad fields. The deployed rule (spread fields, min 8, conf 0.9) makes four wrong calls — 220706 (100% bad, 5 fields) and 230403 (83% bad, 6 fields) called pass with confidence 0.89 and 0.99, 230314 (64% bad) pass at 0.91, 230425 (49% bad) fail at 0.93 — and returns inconclusive on 220606 and 230529. Short chips cannot reach the min-8 guard, so they are the most exposed.

Out-of-distribution detection failed. We tested two standard detectors against the training chips: a Mahalanobis distance on four image statistics (mean, standard deviation, Laplacian focus proxy, dark fraction) and a Mahalanobis distance in the model's 576-dimensional penultimate

feature space. The image-statistics detector flags almost nothing; the feature-space detector flags almost everything (16 of 25 sessions above the training 99th percentile), and **neither flags session 230405**, the worst failure. We therefore do not present an OOD alarm as a safety feature; the distance is exposed as a diagnostic only.

What this means in practice. The tool is usable as a *screening* aid with a human in the loop, not as an autonomous gate. Its measured error rate (13% confident-but-wrong) is the number to quote in any deployment decision.

6.6 Generalisation to a cell line the model has never seen

The session-disjoint protocol of §6.3 keeps all six cell lines in training. To measure a stricter setting we hold out one cell line entirely — test = all of its fields, train = the other five with any session containing the held-out line removed — and retrain the deployed configuration (`audit/05_lolo_cellline.py`):

held-out cell line	test fields (sessions)	accuracy	balanced acc	AUC
A549	775 (24)	0.712	0.661	0.678
CACO	346 (17)	0.512	0.611	0.666
HPMEC	1462 (29)	0.589	0.605	0.659
HUVEC	107 (4)	0.879	0.874	0.930
NHBE	138 (6)	0.601	0.499	0.630
HSAEC	244 (21)	0.730	0.723	0.753
mean of the six	3,072 (101)	0.670	0.662	0.719
in-distribution (all lines seen, §6.3)	684 (25)	0.734	0.733	0.791

Table 8. Leave-one-cell-line-out. The spread between cell lines (AUC 0.630-0.930) is larger than the effect of any modelling choice we tested — a backbone with 2.8× the parameters or 512 px inputs change nothing (§6.3).

Deploying on a new cell line costs about seven AUC points on average and up to 22 accuracy points (Caco-2), and the per-line variation dominates. This is the strictest evaluation we have; we keep the session-disjoint protocol as the headline because it matches how such a tool would be used (a new chip of a known cell line), but any deployment to an unseen line needs re-calibration on that line.

6.7 Label-only simulation overstates deployed performance

A common practice — which we followed first — is to simulate a decision policy on **ground-truth labels**. Our first one (random order, no minimum, all 59 chips) gave 94.1% chip accuracy with 6.6 fields — not comparable with the deployed 82.6%, since rule and chips differ. Like for like, we run the *deployed* rule on the *same* 25 held-out chips with ground-truth field labels and with the model's calls (`audit/label_vs_model_same_rule.py`, Figure 7). Without a minimum, labels give 21/24 = 0.875 with one confident-but-wrong chip and the model 17/24 = 0.708 with six; with the shipped 8-field minimum, labels give 0.875 (8.5 fields) and the model 19/23 = 0.826 (9.5 fields, three confident-but-wrong). Per-field errors (AUC 0.791) let an unguarded rule stop early on a wrong run of calls; the 8-field minimum closes most of that gap.

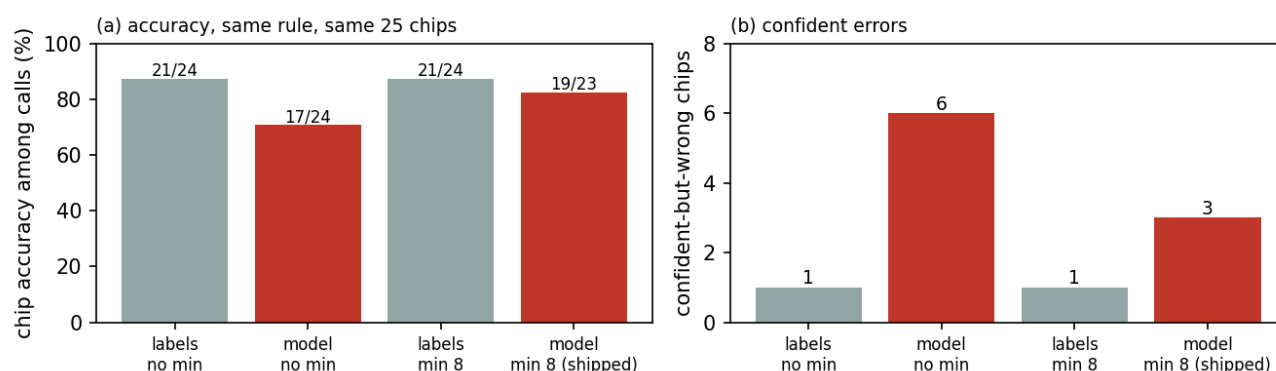


Figure 7. The same rule on the same 25 chips: ground-truth field labels versus model calls.

Recommendation for the field: evaluate QC decision policies **with the model in the loop**. A label-only simulation measures the policy, not the system.

6.8 Interactive demo

A browser-side build of the tool (ONNX Runtime Web, no server) is available at <https://taewoong23-ooc-chip-qc-demo.static.hf.space/index.html>. The bundled example chips use the reference probabilities computed by the Python implementation, so the demo reproduces the numbers in this report exactly; user uploads are scored live in the browser. The page is static, so it stays available throughout the judging period without any server running.

7. Application value

Laboratory automation. The daily QC pass is the most repetitive imaging task in a chip lab: every chip is looked at, most are fine, and the decision is made by eye. The tool returns a ranked plate triage — on 25 test chips it flags 6 failures and defers 2, while spending 238 of 684 fields (65% saved) — so a human looks only where it matters, and the imaging budget follows the risk.

Data assetisation and standardisation. The audit's most transferable result is that a chip imaging dataset's *information content* is not its file count: 3,072 fields carry roughly **181 independent labels**, and a published split that leaks acquisition groups can inflate accuracy by **8.2 pp**. Any organisation building chip data assets — or training models on them — needs exactly this kind of measurement to know what its data is worth and when a reported number can be believed. The corrected protocol (session-level splits, chip-level metrics, explicit sampling policy) is dataset-agnostic and applies to any group-structured imaging corpus.

Drug evaluation and toxicology. QC is the gate in front of every downstream readout. A calibrated, chip-level QC layer — with an explicit *inconclusive* outcome instead of a guess — prevents failing chips from contaminating dose-response curves, and makes the provenance of each excluded chip auditable.

Toward chip digital twins. A digital twin needs a state estimate of the physical system at each time point. The tool's output is exactly that at the culture level: a calibrated pass/fail state plus a spatial map of where the culture is degrading, and a measured cost (9.5 fields) for obtaining it. Feeding such state estimates into a model of the culture over time is the natural next step.

What we do not claim. No wet-lab validation, no biological or clinical validity, one dataset for the tool, and 3 of 23 calls on unseen chips are confident (≥ 0.9) and wrong (§9). The contribution is a trustworthy *measurement and decision layer*, not a biological finding.

8. Discussion

For benchmark authors. The three defects we measure are cheap to check and were not checked here: a session-level (or patient-level, or chip-level) split; a design-effect estimate for the label; and a per-image agreement statistic between a single sample and the group's consensus. We suggest reporting all three alongside the usual accuracy, because they change what the number means. In this benchmark the effective sample size is 6% of the nominal one, and a model trained on the shipped split is evaluated on chips it has already seen.

For laboratories. The practical consequence of clustered failures is that *which* fields are imaged matters: the same rule reads the same chip correctly or not depending on whether it looks at the beginning of the acquisition order or across the whole chip. We could not establish a general ranking between random, scan and adaptive sampling with 59 sessions (§4.3), so the tool uses spread sampling to avoid the specific failure we observed rather than as an optimised policy. Its cost (9.5 fields on average) is modest; its error rate (13% confident-but-wrong) is the number that matters in a deployment decision.

For method developers. The largest single lesson is in Figure 7: without a minimum-field guard the same rule on the same chips is 0.875 accurate on labels and 0.708 with the model in the loop. Per-field errors interact with early stopping. QC papers that report policy numbers from label simulations should be read with this in mind, and we recommend model-in-the-loop evaluation as the default.

Open problem. The dataset's own definition of *bad* mixes technical artifacts with biological failure, which suggests the actionable output should be "re-image" versus "discard". We tried and failed to validate that mapping (§4.5): the causal question cannot be observed without actual re-imaging. A dataset that images each chip twice — before and after a re-imaging attempt — would settle it, and would be a small, valuable addition to this benchmark.

9. Limitations

1. **13% confident-but-wrong** on unseen chips: 4 of 23 calls are wrong, 3 of them with ≥ 0.9 confidence. This is a research prototype, not a validated instrument; a chip should not be discarded on its output alone.
2. **No reliable out-of-distribution detector.** We tested image-statistics distance and feature-space Mahalanobis distance against the training chips; both failed to flag the worst case (the 100%-bad chip called *pass* with 0.94 confidence). The tool exposes the distance as a diagnostic only.
3. **The re-image/discard recommendation is not validated** (§4.5) and is not part of the tool's output.
4. **One dataset, 25 test chips.** Chip-level metrics carry wide confidence intervals; we report them as point estimates with the split size. No wet-lab validation was performed, and we make no biological or clinical claim — the claim is about *measurement validity*.
5. **Sampling policies could not be ranked.** The label-based comparison (§4.3, Table 3) is underpowered when repeated with the model in the loop (13–15 chips; all paired CIs

include zero). We report the policy we use and the failure that motivates it, not a ranking.

6. **Labels come from a four-rater majority**, so the ceiling of any model on this benchmark is the agreement among experts, which is not public. Our numbers are therefore a lower bound in that specific sense, while the leakage inflation is an upper-bound problem.
7. **Generalisation to a new cell line is untested in deployment**. Held-out cell lines cost ~ 7 AUC points on average ($0.791 \rightarrow 0.719$) and the spread between lines dominates every modelling choice we tested (§6.6). The tool is validated for cell lines it has seen.
8. **The acquisition order is used as a proxy for spatial order**. Consecutive fields are highly correlated ($r = 0.63$ vs 0.09 – 0.38 for random pairs), which supports the proxy, but the exact stage geometry is not public.

10. Reproducibility

```
pip install -r requirements.txt          # torch, torchvision, numpy, Pillow, scipy,
scikit-learn,                           # matplotlib, openpyxl, gradio, markdown,
playwright

# data (not redistributed here) – the zip holds a top-level OOC_image_dataset/ folder
mkdir -p ../data
curl -L -o ooc.zip
"https://zenodo.org/api/records/10203721/files/OOC_image_dataset.zip/content"
python3 -c "import zipfile; zipfile.ZipFile('ooc.zip').extractall('../data')" #
unzip(1) fails on this zip64
rm ooc.zip                               # lands at ../data/OOC_image_dataset/ (or set
OOO_DATA)

python3 audit/leakage_controlled.py      # +8.2 pp / +9.6 pp (finding 1, 8 seeds)
python3 audit/block_structure.py         # ICC 0.321, effective N  $\approx$  181 (finding 2)
python3 audit/structure_probe.py         # runs test (finding 3)
python3 audit/adaptive_sampling.py       # policy comparison (finding 4)
python3 audit/recovery_test.py           # negative result (finding 5)
python3 evaluate.py                     # deployed-tool numbers (§6.3)
python3 figures.py                      # every figure in this report
python3 demo/app.py                     # interactive demo
```

Every number in this report is written to `results/*.json` by the script that produced it.

11. Related work

Leakage in benchmarks. Split leakage is a known and actively studied class of problem: Ramos et al. audit leakage across seven visual datasets and measure its effect on downstream evaluation [7]; whole-slide-image multimodal benchmarks have been shown to leak at patient and institution level, with a measurable accuracy gap between leaked and audit-clean cases [8]; OCT classification was shown to be inflated by improper splitting [9]. Tooling exists for near-duplicate detection and grouped splitting [10,11]. Our finding is a new instance of this class, with the magnitude measured in an OoC QC benchmark by a controlled A/B.

Clustered labels and effective sample size. Treating images as correlated clusters when computing power or sample size is standard in imaging study design [12], and effective sample size under autocorrelation is textbook. We apply these tools to field-level QC labels and quantify the consequence for a benchmark (≈ 181 independent labels).

Microscopy QC. Automated QC for high-throughput microscopy has its own benchmarks and methods (e.g. AutoQC-Bench, 2025 [13]); classical QC metrics for microscopy images are long established [14]. Our contribution is complementary: not a new QC metric, but a measurement audit and a sampling/decision policy for chip-level QC.

Organoid/OoC imaging benchmarks. Several organoid imaging datasets exist (detection, segmentation, tracking), including multi-rater annotation [15]. None, to our knowledge, ship a chip-level QC protocol; the field-dependence we measure is a property of this label definition.

12. Conclusion

A public benchmark for organ-on-a-chip quality control reports per-image accuracy on a split that leaks sessions (+8.2 pp measured by controlled A/B, 8 seeds), with labels that are clustered along the acquisition order (ICC 0.321; effective $N \approx 181$ of 3,072) and field-dependent (a single field agrees with the session majority 78.8% of the time). Correcting the protocol — session-grouped splits, chip-level evaluation, explicit sampling policy — and sampling fields spread across the chip yields a tool that reaches 82.6% chip-level accuracy among confident calls with 9.5 fields on average, and says "inconclusive" when it cannot decide. Two negative results bound the claims: the re-image/discard distinction is not supported by this data, and no reliable OOD detector was found. Simulating the same stopping rule on ground-truth labels overstates its performance with the model in the loop (0.875 vs 0.708 without a minimum-field guard), a caution that generalises beyond this benchmark.

Future work. (i) Extend the model-in-the-loop versus label-only comparison to a second dataset and modality (we have begun with a patient-derived-organoid drug-response dataset); (ii) a calibrated stopping rule that accounts for the measured design effect; (iii) prospective validation of the two-mode (transient vs persistent) hypothesis with actual re-imaging.

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Code licence MIT; derived artefacts CC-BY-SA; dataset images not redistributed beyond the 44 attributed demo example fields.